

Shubh Gupta

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EDUCATION

Vellore Institute of Technology

Integrated M.Tech in Artificial Intelligence

CGPA: 8.73

October 2022 – May 2027

INTERNSHIP EXPERIENCE

AI Intern – Information Technology Department

MP Birla Group (Birla Corporation Limited)

Satna, Madhya Pradesh

October 2025 – Dec 2025

- Developed an AI-based OCR automation system to digitize organizational forms, ensuring data protection and infrastructure security in alignment with industry best practices.
- Implemented image preprocessing and OCR pipelines using Python, OpenCV, and Tesseract, conducting vulnerability testing and risk analysis to ensure secure data handling.
- Achieved **92% extraction accuracy** while reducing manual data-entry effort, collaborating with cross-functional teams to monitor system performance and support incident response activities.

PROJECTS

Deep Image Steganography | Python, TensorFlow, PyTorch, OpenCV | GitHub | Live Demo

April 2025

- Designed GAN, VAE, and DDPM-based models to securely embed and recover images with minimal distortion, implementing cyber security measures to protect sensitive data and enable covert communication.
- Built encoder-decoder and diffusion pipelines for high-fidelity image reconstruction, conducting vulnerability testing to ensure robustness against data extraction attacks and maintaining data integrity.
- Improved robustness by **18%** using adversarial loss tuning and diffusion-based denoising, collaborating with team members in a dynamic environment to explore secure information transfer applications.

FaceGuard AI | FastAPI, YOLOv8, FaceNet, React.js, WebSockets | GitHub

Sep 2025

- Developed a real-time face recognition system with live video streaming using FastAPI and WebSockets, implementing information security assessments to protect user data and ensure compliance with security best practices.
- Implemented YOLOv8 for face detection and FaceNet with one-shot learning for accurate user identification, conducting risk analysis and vulnerability testing to strengthen system resilience against unauthorized access.
- Reduced face-matching latency by **40%** using FAISS-based vector search and asynchronous inference optimization, collaborating with teams to monitor threats and support incident response activities.

Verba - Document Q&A Assistant | Python, Django, RAG, Gemini, Groq | GitHub | Live Demo

Nov 2025

- Built a full-stack RAG-powered document Q&A system supporting PDF, DOCX, and TXT file uploads, implementing data protection measures and conducting compliance reviews to ensure secure handling of sensitive information.
- Implemented document chunking, embedding generation with Gemini API, and vector storage using ChromaDB for efficient retrieval, applying security best practices to protect data infrastructure and maintain information security.
- Integrated Groq Llama for context-aware answer generation with session-based chat history and responsive UI, demonstrating problem-solving skills and ability to multitask in building secure applications.

SKILLS

Programming Languages: Python, C++, SQL

Artificial Intelligence: Machine Learning, Deep Learning, NLP, Computer Vision

Frameworks & Tools: TensorFlow, PyTorch, Keras, OpenCV, GitHub, Google Colab, VS Code, IntelliJ

Security & Infrastructure: Vulnerability Testing, Risk Analysis, Data Protection, Compliance Reviews

ACHIEVEMENTS

- **NASSCOM AI Hackathon:** Semi-finalist
- **TATA Imagination Challenge 2025:** Finalist
- **Conference Paper:** In preparation on AI/ML applications